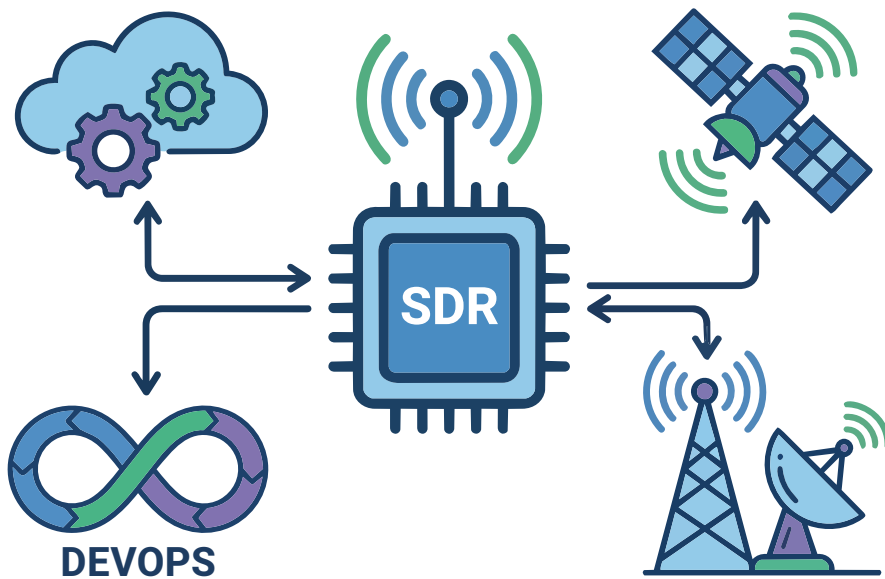


DevOps in Software Defined Radio: The Future of Wireless and Telecom

The integration of DevOps in radio and wireless applications can help automate infrastructure deployments in sensitive domains like the military.



Software Defined Radio (SDR) refers to the advanced radio communication technology in which radio and wireless infrastructure are managed, controlled, allocated and monitored by software libraries for real-time high performance applications. In traditional radio engineering applications, hard-wired infrastructure (including circuits, resistors, capacitors, antenna and similar components) is used and managed manually. SDR integrates the mechanisms to control the entire radio, wireless and telecom environment by software applications locally or remotely.

For high performance and security-based applications for air defence, military, weather satellites and public safety, SDR-based digital environments

use field programmable gate arrays (FPGAs) for accurate results without unpredictable delays.

Key applications of SDR are:

- Military and defence
- Telecom and spectrum analytics
- Satellites and navigation systems
- Industrial monitoring/SCADA
- Wireless environment testing and measurements
- Global navigation satellite systems (GNSS)
- Cybersecurity and signals intelligence
- Aviation and maritime tracking
- Industrial IoT and smart grid
- Wireless protocol research
- Wireless protocol reverse engineering
- Emergency and public safety

DevOps in SDR

DevOps can be used in multiple domains including civil and construction engineering, radio frequency and telecom, wireless spectrum analysis and dynamic allocations, electronic components assembly, mechanical engineering, etc (Figure 1).

Telecom and wireless technologies can use DevOps for taking specific and error-free actions on air waves or radio signals, which can be managed, controlled and monitored using SDR. The integration of SDR with DevOps makes the overall radio and telecom environment secure and privacy-aware.

Here's the pipeline/workflow of RF and DevOps:

Commit -> Build -> Simulate -> Test (HIL) -> Deploy